

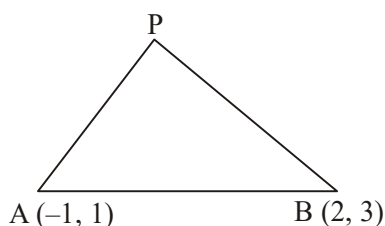
21. Let A(-1, 1) and B(2, 3) be two points and P be a variable point above the line AB such that the area of $\triangle PAB$ is 10. If the locus of P is $ax + by = 15$, then $5a + 2b$ is :

माना दो बिन्दु A(-1, 1) तथा B(2, 3) हैं और रेखा AB के ऊपर एक चर बिन्दु P इस प्रकार है कि $\triangle PAB$ का क्षेत्रफल 10 है। यदि P का बिंदुपथ $ax + by = 15$ है, तो $5a + 2b$ बराबर है :

- (A) 6 (B) 4 (C) $-\frac{6}{5}$ (D) $-\frac{12}{5}$

Ans. D

Sol.



Locus of P,

$$ax + by = 15$$

Take point P as (h, k)

Area of $\triangle ABP$

$$= \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)| = 10$$

$$= \frac{1}{2} |-1(3-k) + 2(k-1) + h(1-3)| = 10$$

$$= |-3 + k + 2k - 2 + h - 3h| = 20$$

$$= |3k - 2h - 5| = 20$$

$$= -(3k - 2h - 5) = 20 \Rightarrow 2h - 3k = 15$$

$$a = 2, b = -3$$

$$\therefore 5a + 2b = 4$$

22. If the line segment joining the points (5,2) and (2,a) subtends an angle $\frac{\pi}{4}$ at the origin, then the absolute value of the product of all possible values of a is :

यदि बिन्दुओं (5,2) तथा (2,a) को मिलाने वाला रेखाखंड, मूलबिन्दु पर $\frac{\pi}{4}$ का कोण बनाता है, तो a के सभी संभव मानों के गुणनफल का निरपेक्ष मान है :

(A) 2

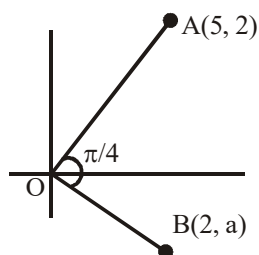
(B) 8

(C) 4

(D) 6

Ans. C

Sol.



$$M_{OA} = \frac{2}{5}$$

$$M_{OB} = \frac{a}{2}$$

$$= \tan \frac{\pi}{4} = \left| \frac{\frac{2}{5} - \frac{a}{2}}{1 + \frac{2}{5} \cdot \frac{a}{2}} \right|$$

$$\Rightarrow 1 = \left| \frac{4 - 5a}{10 + 2a} \right| \Rightarrow a = \frac{-6}{7}, \frac{14}{3}$$

Product $\Rightarrow -4$, Abs. value = 4

23. A ray of light coming from the point P(1, 2) gets reflected from the point Q on the x-axis and then passes through the point R(4, 3). If the point S(h, k) is such that PQRS is a parallelogram, then hk^2 is equal to :

बिन्दु P(1, 2) से आने वाली रोशनी की एक किरण x-अक्ष पर बिन्दु Q से परावर्तित होने के पश्चात् बिन्दु R(4, 3) से गुजरती है। यदि बिन्दु S(h, k) इस प्रकार है कि PQRS एक समांतर चतुर्भुज है, तो hk^2 बराबर है :

(A) 80

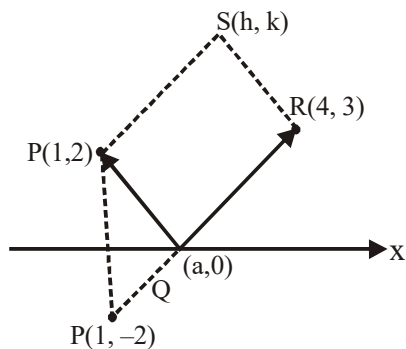
(B) 90

(C) 60

(D) 70

Ans. D

Sol.



$$P'R : y + 2 = \frac{5}{3}(x - 1)$$

For Point Q $\Rightarrow y = 0$

$$\frac{6}{5} = a - 1 \Rightarrow a = \frac{11}{5}$$

Now, PQRS is parallelogram

$$\therefore \frac{h+a}{2} = \frac{4+1}{2} \Rightarrow h = 5 - \frac{11}{5} = \frac{14}{5}$$

$$\text{and } \frac{2+3}{2} = \frac{k}{2} \Rightarrow K = 5$$

$$\text{Now } hk^2 = 25 \times \frac{14}{5} = 14 \times 5 = 70$$

24. Two vertices of a triangle ABC are A(3,-1) and B(-2,3), and its orthocentre is P(1,1). If the coordinates of the point C are (α , β) and the centre of the circle circumscribing the triangle PAB is (h, k), then the value of ($\alpha + \beta$) + 2(h + k) equals :

किसी त्रिभुज ABC के दो शीर्ष A(3,-1) और B(-2,3) हैं तथा उसका लंब केन्द्र P(1,1) है। यदि बिन्दु C के निर्देशांक (α , β) हैं तथा त्रिभुज PAB के परिवृत्त का केन्द्र (h, k) है, तब ($\alpha + \beta$) + 2(h + k) का मान बराबर है :

(A) 5

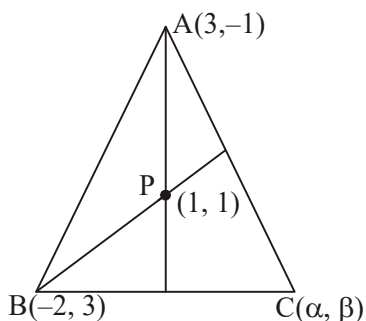
(B) 51

(C) 15

(D) 81

Ans. A

Sol.



$$\text{Equation of PC : } y - 1 = \frac{5}{4}(x - 1) \left(\because m_{AB} = -\frac{4}{5} \right)$$

$$4y - 4 = 5x - 5$$

$$5x - 4y = 1$$

_____ (1)

$$\text{Equation of BC : } y - 3 = 1(x + 2) \left(\because m_{AP} = -1 \right)$$

$$x - y = -5$$

_____ (2)

Solving equation (1) and (2), $x = 21$, $y = 26$

$$\therefore \alpha = 21, \beta = 26 \Rightarrow (\alpha + \beta) = 47$$

$$\text{Now, equation of } \perp \text{ bisector of AP : } y - 0 = 1(x - 2)$$

$$x - y = 2 \quad \text{_____ (3)}$$

$$\text{Equation of } \perp \text{ bisector of AB : } y - 1 = \frac{5}{4} \left(x - \frac{1}{2} \right) \quad \text{_____ (4)}$$

Solving equation (3) and (4),

$$x = -\frac{19}{2}, y = -\frac{23}{2}$$

$$\text{i.e } 2h = -19, 2k = -23$$

$$\therefore 2(h + k) = -42$$

$$\text{So, } (\alpha + \beta) + 2(h + k) = 47 - 42 = 5$$

25. The portion of the line $4x + 5y = 20$ in the first quadrant is trisected by the lines L_1 and L_2 passing through the origin.

The tangent of an angle between the lines L_1 and L_2 is :

रेखा $4x + 5y = 20$ के प्रथम चतुर्थांश में भाग को मूल बिन्दु से होकर जाने वाली रेखाएँ L_1 तथा L_2 समत्रिभाजित करती हैं।

L_1 तथा L_2 के बीच एक कोण की स्पर्शज्या है :

(A) $\frac{8}{5}$ (B) $\frac{2}{5}$ (C) $\frac{30}{41}$ (D) $\frac{25}{41}$

Ans. C

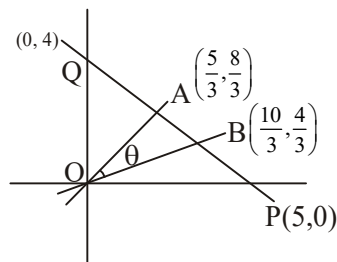
Sol. $A\left(\frac{5}{3}, \frac{8}{3}\right), B\left(\frac{10}{3}, \frac{4}{3}\right)$

$$m_{OA} = m_2 = \frac{8}{5}$$

$$m_{OB} = m_1 = \frac{2}{5}$$

$$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right| = \left| \frac{\frac{8}{5} - \frac{2}{5}}{1 + \frac{8}{5} \times \frac{2}{5}} \right|$$

$$= \frac{\frac{6}{5}}{1 + \frac{16}{25}} = \frac{6 \times 5}{25 + 16} = \frac{30}{41}$$



26. Let R be the interior region between the lines $3x - y + 1 = 0$ and $x + 2y - 5 = 0$ containing the origin. The set of all values of a, for which the points $(a^2, a + 1)$ lie in R, is :

माना रेखाओं $3x - y + 1 = 0$ तथा $x + 2y - 5 = 0$ के अंदर का क्षेत्र, जिसमें मूलबिन्दु है, R है। a के सभी मानों, जिनके लिए

$(a^2, a+1)$ क्षेत्र R में है, का समुच्चय है :

(A) $(-3, -1) \cup \left(-\frac{1}{3}, 1\right)$

(B) $(-3, -1) \cup \left(\frac{1}{3}, 1\right)$

(C) $(-3, 0) \cup \left(\frac{1}{3}, 1\right)$

(D) $(-3, 0) \cup \left(\frac{2}{3}, 1\right)$

Ans. C

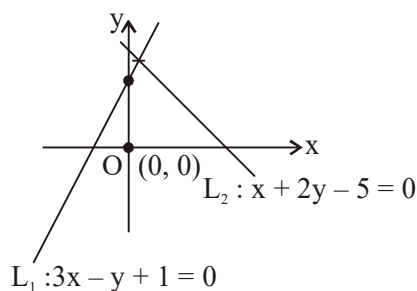
Sol. $P(a^2, a+1)$

$$L_1 = 3x - y + 1 = 0$$

Origin and P lies same side w.r.t. L_1

$$\Rightarrow L_1(0) \cdot L_1(P) > 0$$

$$\therefore 3(a^2) - (a+1) + 1 > 0$$



$$\Rightarrow 3a^2 - a > 0$$

$$a \in (-\infty, 0) \cup \left(\frac{1}{3}, \infty\right) \quad \text{_____ (1)}$$

Let $L_2 : x + 2y - 5 = 0$

Origin and P lies same side w.r.t. L_2

$$\Rightarrow L_2(0) \cdot L_2(P) > 0$$

$$\Rightarrow a^2 + 2(a+1) - 5 < 0$$

$$\Rightarrow a^2 + 2a - 3 < 0$$

$$\Rightarrow (a+3)(a-1) < 0$$

$$\therefore a \in (-3, 1) \quad \text{_____ (2)}$$

Intersection of (1) and (2)

$$a \in (-3, 0) \cup \left(\frac{1}{3}, 1\right)$$

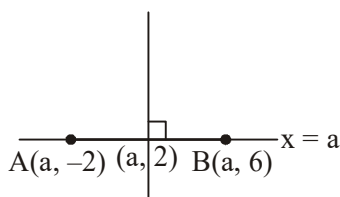
27. Let $\left(5, \frac{a}{4}\right)$ be the circumcenter of a triangle with vertices $A(a, -2)$, $B(a, 6)$ and $C\left(\frac{a}{4}, -2\right)$. Let α denote the circumradius, β denote the area and γ denote the perimeter of the triangle. Then $\alpha + \beta + \gamma$ is :

माना $A(a, -2)$, $B(a, 6)$ तथा $C\left(\frac{a}{4}, -2\right)$ के एक त्रिभुज का परिकेन्द्र $\left(5, \frac{a}{4}\right)$ है। यदि इस त्रिभुज के परिवृत्त की त्रिज्या α , त्रिभुज का क्षेत्रफल β तथा त्रिभुज का परिमाप γ है, तो $\alpha + \beta + \gamma$ is बराबर है :

- (A) 53 (B) 60 (C) 62 (D) 30

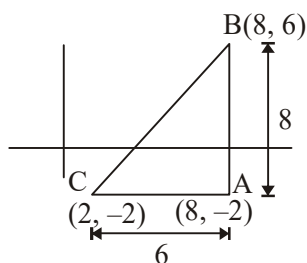
Ans. A

Sol.



Circumcentre lie on perpendicular bisector of $AB \Rightarrow y = 2$

$$\Rightarrow \frac{a}{4} = 2 \quad \Rightarrow a = 8$$



$$\Rightarrow \text{Perimeter} = 6 + 8 + 10 = 24$$

$$\text{Area} = \frac{6 \times 8}{2} = 24$$

Circumradius = half of hypotenuse

$$= \frac{10}{2} = 5$$

$$\alpha + \beta + \gamma = 24 + 24 + 5 = 53$$

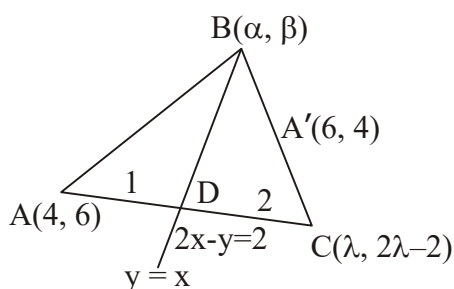
28. In a $\triangle ABC$, suppose $y = x$ is the equation of the bisector of the angle B and the equation of the side AC is $2x - y = 2$. If $2AB = BC$ and the points A and B are respectively $(4, 6)$ and (α, β) , then $\alpha + 2\beta$ is equal to :

एक $\triangle ABC$ में माना कोण B के समद्विभाजक का समीकरण $y = x$ है तथा भुजा AC का समीकरण $2x - y = 2$ है। यदि (4, 6) और (α, β) क्रमशः बिन्दु A और B तथा $2AB = BC$ है, तो $\alpha + 2\beta$ बराबर है

- (A) 48 (B) 39 (C) 45 (D) 42

Ans. D

Sol.



$$\frac{AB}{BC} = \frac{AD}{DC} = \frac{1}{2}$$

$$D\left(\frac{\lambda + 8}{3}, \frac{2\lambda - 2 + 12}{3}\right) \text{ lies on } y = x$$

$$\frac{\lambda + 8}{3} = \frac{2\lambda + 10}{3}$$

$$\Rightarrow \lambda = -2 \Rightarrow c(-2, -6)$$

Image of A w.r.t. $y = x$ lies on BC

$$m(BC) = m(A'C)$$

$$\Rightarrow \frac{\alpha + 6}{\alpha + 2} = \frac{4 + 6}{6 + 2} \Rightarrow \alpha = 14 = \beta$$

$$\alpha + 2\beta = 42$$

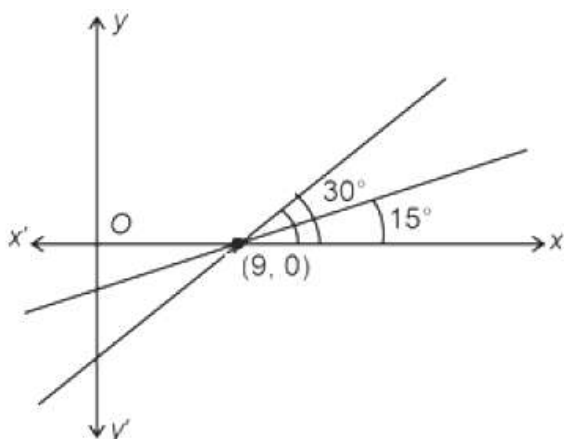
29. A line passing through the point A (9, 0) makes an angle of 30° with the positive direction of x-axis. If this line is rotated about A through an angle of 15° in the clockwise direction, then its equation in the new position is :

बिंदु A (9, 0) से होकर जाने वाली एक रेखा x-अक्ष की धनात्मक दिशा से 30° का कोण बनाती है। यदि इस रेखा को घड़ी की दिशा में 15° के कोण तक A के सापेक्ष घुमाया जाता है, तो नयी स्थिति में इसका समीकरण है :

- (A) $\frac{x}{\sqrt{3}-2} + y = 9$ (B) $\frac{x}{\sqrt{3}+2} + y = 9$ (C) $\frac{y}{\sqrt{3}-2} + x = 9$ (D) $\frac{y}{\sqrt{3}+2} + x = 9$

Ans. C

Sol.



Equation : $y - 0 = \tan 15^\circ (x - 9) \Rightarrow y = (2 - \sqrt{3})(x - 9)$

30. If $x^2 - y^2 + 2hxy + 2gx + 2fy + c = 0$ is the locus of a point, which moves such that it is always equidistant from the lines $x + 2y + 7 = 0$ and $2x - y + 8 = 0$, then the value of $g + c + h - f$ equals :

यदि एक बिन्दु, जो इस प्रकार चलता है कि इसकी रेखाओं $x + 2y + 7 = 0$ तथा $2x - y + 8 = 0$ से दूरी बराबर रहती है, का बिन्दुपथ $x^2 - y^2 + 2hxy + 2gx + 2fy + c = 0$ है, तो $g + c + h - f$ का मान बराबर है :

- (A) 6 (2) 8 (C) 29 (D) 14

Ans. D

Sol. Locus of point $P(x, y)$ whose distance from
Gives

$x + 2y + 7 = 0$ & $2x - y + 8 = 0$ are equal is $\frac{x + 2y + 7}{\sqrt{5}} = \pm \frac{2x - y + 8}{\sqrt{5}}$

$(x + 2y + 7)^2 - (2x - y + 8)^2 = 0$

Combined equation of lines

$(x - 3y + 1)(3x + y + 15) = 0$

$3x^2 - 3y^2 - 8xy + 18x - 44y + 15 = 0$

$x^2 - y^2 - \frac{8}{3}xy + 6x - \frac{44}{3}y + 5 = 0$

$x^2 - y^2 + 2hxy + 2gx + 2fy + c = 0$

$h = \frac{4}{3}, g = 3, f = -\frac{22}{3}, c = 5$

$g + c + h - f = 3 + 5 - \frac{4}{3} + \frac{22}{3} = 8 + 6 = 14$

31. Let $\alpha, \beta, \gamma, \delta \in \mathbb{Z}$ and let $A(\alpha, \beta), B(1, 0), C(\gamma, \delta)$ and $D(1, 2)$ be the vertices of a parallelogram ABCD. If

$AB = \sqrt{10}$ and the points A and C lie on the line $3y = 2x + 1$, then $2(\alpha + \beta + \gamma + \delta)$ is equal to :

माना $\alpha, \beta, \gamma, \delta \in \mathbb{Z}$ हैं तथा माना एक समांतर चतुर्भुज ABCD के शीर्ष $A(\alpha, \beta), B(1,0), C(\gamma, \delta)$ तथा $D(1,2)$ हैं। यदि $AB = \sqrt{10}$ है तथा बिन्दु A और C रेखा $3y = 2x + 1$ पर हैं, तो $2(\alpha + \beta + \gamma + \delta)$ बराबर है:

- (A) 12 (B) 10 (C) 5 (D) 8

Ans. D

Sol. ABCD is a parallelogram so mid-point of AC = mid-point of BD

$$\Rightarrow \left(\frac{\alpha + \gamma}{2}, \frac{\beta + \delta}{2} \right) = \left(\frac{1+1}{2}, \frac{0+2}{2} \right)$$

$$\Rightarrow \left(\frac{\alpha + \gamma}{2}, \frac{\beta + \delta}{2} \right) = (1, 1)$$

$$\Rightarrow \alpha + \gamma = 2 \text{ and } \beta + \delta = 2$$

$$\text{So, } 2(\alpha + \beta + \gamma + \delta) = 2(2 + 2) = 8$$

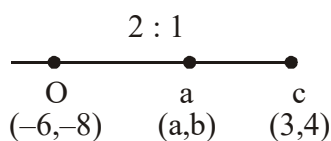
32. Let $A(a, b), B(3, 4)$ and $C(-6, -8)$ respectively denote the centroid, circumcentre and orthocentre of a triangle. Then, the distance of the point $P(2a + 3, 7b + 5)$ from the line $2x + 3y - 4 = 0$ measured parallel to the line $x - 2y - 1 = 0$ is :

माना $A(a, b), B(3, 4)$ तथा $C(-6, -8)$ एक त्रिभुज के केन्द्रक, परिकेन्द्र तथा लंबकेन्द्र हैं। तो बिन्दु $P(2a + 3, 7b + 5)$ की रेखा $2x + 3y - 4 = 0$ से, रेखा $x - 2y - 1 = 0$ के समांतर नापी गई दूरी है :

- (A) $\frac{17\sqrt{5}}{6}$ (B) $\frac{17\sqrt{5}}{7}$ (C) $\frac{\sqrt{5}}{17}$ (D) $\frac{15\sqrt{5}}{7}$

Ans. B

Sol. Centroid divides orthocentre and circumcentre in 2 : 1.

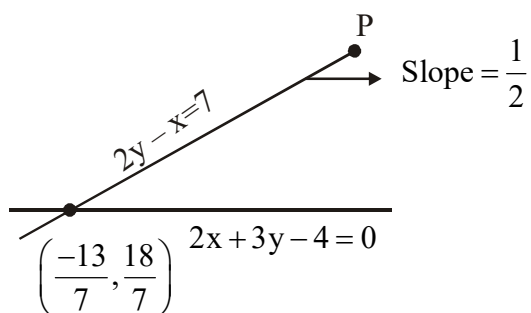


$$a \equiv \frac{2(3) - 6}{3} = 0$$

$$b \equiv \frac{2(4) - 8}{3} = 0$$

$$\Rightarrow (a, b) \equiv (0, 0)$$

$$\Rightarrow P \equiv (3, 5)$$



$$\Rightarrow (y - 5) = \frac{1}{2}(x - 3)$$

$$\Rightarrow 2y - 10 = x - 3$$

$$\Rightarrow 2y - x = 7$$

$$\Rightarrow \sqrt{\left(3 + \frac{13}{7}\right)^2 + \left(5 - \frac{18}{7}\right)^2} = \frac{\sqrt{34^2 + 17^2}}{7}$$

$$= \frac{17\sqrt{5}}{7}$$

33. If the sum of squares of all real values of α , for which the lines $2x - y + 3 = 0$, $6x + 3y + 1 = 0$ and $\alpha x + 2y - 2 = 0$ do not form a triangle is p , then the greatest integer less than or equal to p is _____.

माना α के सभी वास्तविक मानों, जिनके लिए रेखाएँ $2x - y + 3 = 0$, $6x + 3y + 1 = 0$ तथा $\alpha x + 2y - 2 = 0$ एक त्रिभुज नहीं बनाती हैं, के वर्गों का योग p है, तो महत्तम पूर्णांक $\leq p$ है _____

Ans. 32

Sol. $2x - y + 3 = 0$

$$6x + 3y + 1 = 0$$

$$\alpha x + 2y - 2 = 0$$

will not form a Δ if $\alpha x + 2y - 2 = 0$ is concurrent with $2x - y + 3 = 0$ and $6x + 3y + 1 = 0$ or parallel to either of them so

Case-1: Concurrent lines

$$\begin{vmatrix} 2 & -1 & 3 \\ 6 & 3 & 1 \\ \alpha & 2 & -2 \end{vmatrix} = 0 \Rightarrow \alpha = \frac{4}{5}$$

Case-2 : Parallel lines

$$-\frac{\alpha}{2} = \frac{-6}{3} \text{ or } -\frac{\alpha}{2} = 2$$

$$\Rightarrow \alpha = 4 \text{ or } \Rightarrow \alpha = -4$$

$$P = 16 + 16 + \frac{16}{25}$$

$$[P] = \left[32 + \frac{16}{25} \right] = 32$$

34. Let $A(6, 8)$, $B(10 \cos \alpha, -10 \sin \alpha)$ and $C(-10 \sin \alpha, 10 \cos \alpha)$, be the vertices of a triangle. If $L(a, 9)$ and $G(h, k)$ be its orthocenter and centroid respectively, then $(5a - 3h + 6k + 100 \sin 2\alpha)$ is equal to _____.

माना कि $A(6, 8)$, $B(10 \cos \alpha, -10 \sin \alpha)$ और $C(-10 \sin \alpha, 10 \cos \alpha)$ एक त्रिभुज के शीर्ष हैं। यदि $L(a, 9)$ और $G(h, k)$ इसके क्रमशः लम्बकेन्द्र और केन्द्रक हैं, तो $(5a - 3h + 6k + 100 \sin 2\alpha)$ बराबर है _____।

Ans. 145

Sol. $A(6, 8)$

$B(10 \cos \theta, -10 \sin \alpha)$

$C(-10 \sin \alpha, 10 \cos \alpha)$

Lies on circle $x^2 + y^2 = 10^2$

so circumcentre $(C) = (0, 0)$

$$\begin{array}{c} \text{2 : 1} \\ \text{O(a, 9) \quad G(h, k) \quad C(0, 0)} \end{array}$$

$$\frac{O+a}{3} = h \quad \frac{9+0}{3} = k$$

$$a = 3h \quad k = 3$$

$$\text{Now } h = \frac{6 + 10 \cos \alpha - 10 \sin \alpha}{3}$$

$$k = 3 = \frac{8 - 10 \sin \alpha + 10 \cos \alpha}{3}$$

$$3h = 6 + 10 \cos \alpha - 10 \sin \alpha$$

$$10 \cos \alpha - \sin \alpha = 1$$

$$3h = 6 + 1$$

$$h = 7/3 \quad 3h = 7 \quad a = 7$$

$$1 = 10(\cos \alpha - \sin \alpha)$$

square

$$1 = 100(1 - \sin 2\alpha)$$

$$100 \sin 2\alpha = 99$$

$$\text{Now } 5a - 3h + 6k + 100 \sin 2\alpha$$

$$\Rightarrow 35 - 7 + 18 + 99$$

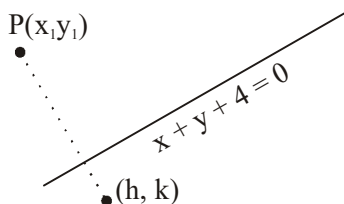
$$= 145$$

35. Let A and B be the two points of intersection of the line $y + 5 = 0$ and the mirror image of the parabola $y^2 = 4x$ with respect to the line $x + y + 4 = 0$. If d denotes the distance between A and B, and a denotes the area of ΔSAB , where S is the focus of the parabola $y^2 = 4x$, then the value of $(a + d)$ is _____.

माना कि परवलय $y^2 = 4x$ के रेखा $x + y + 4 = 0$ में दर्पण प्रतिबिम्ब और रेखा $y + 5 = 0$ के प्रतिच्छेद बिन्दु A और B हैं। यदि A और B हैं। यदि A और B के बीच की दूरी d है और ΔSAB का क्षेत्रफल a है, जहाँ S परवलय $y^2 = 4x$ की नाभि है, तो $(a + d)$ का मान है _____

Ans. 14

Sol. Mirror image of $y^2 = 4x$
with respect to $x + y + 4 = 0$



$$\Rightarrow \left(\frac{h - x_1}{1} \right) = \left(\frac{k - y_1}{1} \right) = -\frac{2(x_1 + y_1 + 4)}{1^2 + 1^2}$$

$$\Rightarrow h - x_1 = k - y_1 = -x_1 - y_1 - 4$$

$$h = -y_1 - 4 \Rightarrow -(h + 4) = y_1$$

$$k = -x_1 - 4 \Rightarrow -(k + 4) = x_1$$

$$y_1^2 = 4x_1$$

$$(-(h + 4))^2 = -4k - 16$$

$$\Rightarrow h^2 + 8h + 4k + 32 = 0$$

$$\Rightarrow x^2 + 8x + 4y + 32 = 0$$

Intersection with $(y + 5 = 0)$

$$y = -5$$

$$x^2 + 8x + 4(-5) + 32 = 0$$

$$x^2 + 8x + 12 = 0$$

$$(x + 6)(x + 2) = 0$$

$$x = -6 \text{ \& } x = -2$$

$$(-6, -5) \quad (-2, -5)$$

Point A Point B

d = Distance between A & B

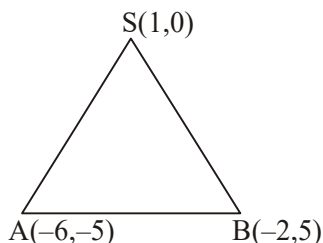
$$d = 4$$

$$a = \text{ar}(\Delta SAB)$$

(Equation of mirror image of $y^2 = 4x$ w.r.t. $x + y + 4 = 0$)

$$(1, 0)$$

Point S (Focus of $y^2 = 4x$)



$$a = \frac{1}{2} \begin{vmatrix} 1 & 0 & 1 \\ -6 & -5 & 1 \\ -2 & -5 & 1 \end{vmatrix}$$

$$R_2 \rightarrow R_2 - R_3$$

$$a = \frac{1}{2} \begin{vmatrix} 1 & 0 & 1 \\ -4 & 0 & 0 \\ -2 & -5 & 1 \end{vmatrix} = \left(\frac{1}{2}\right)(4)(5)$$

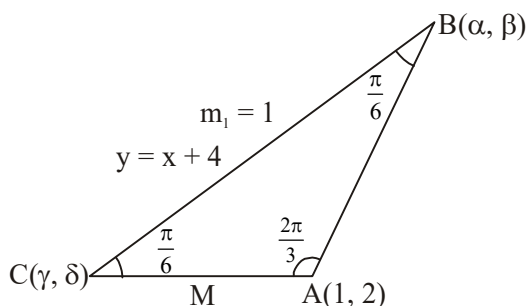
$$\Rightarrow (a + d) = 10 + 4 = 14$$

36. Consider a triangle ABC having the vertices A(1,2), B(α , β) and C(γ , δ) and angles $\angle ABC = \frac{\pi}{6}$ and $\angle BAC = \frac{2\pi}{3}$. If the points B and C lie on the line $y = x + 4$, then $\alpha^2 + \gamma^2$ is equal to _____.

एक त्रिभुज ABC जिसके शीर्ष A(1,2), B(α , β) तथा C(γ , δ) और कोण $\angle ABC = \frac{\pi}{6}$ तथा $\angle BAC = \frac{2\pi}{3}$ हैं पर विचार कीजिए। यदि बिन्दु B तथा C रेखा $y = x + 4$ पर स्थित हैं, तो $\alpha^2 + \gamma^2$ बराबर है _____

Ans. 14

Sol.



Equation of line passing through P(1, 2) and making angle $\frac{\pi}{6}$ with $y = x + 4$ _____ (1)

$$y - 2 = m(x - 1)$$

$$y - 2 = (2 \pm \sqrt{3})(x - 1) \quad \text{_____ (2)}$$

Solve (1) and (2)

$$x + 4 - 2 = (2 \pm \sqrt{3})(x - 1)$$

$$\tan \frac{\pi}{6} = \left| \frac{m - m_1}{1 + mm_1} \right| \pm \frac{1}{\sqrt{3}} = \frac{m - 1}{1 + m}$$

$$m = 2 \pm \sqrt{3}$$

$$x = \frac{4 + \sqrt{3}}{1 + \sqrt{3}} = \alpha \text{ and } x = \frac{4 - \sqrt{3}}{1 - \sqrt{3}} = \gamma$$

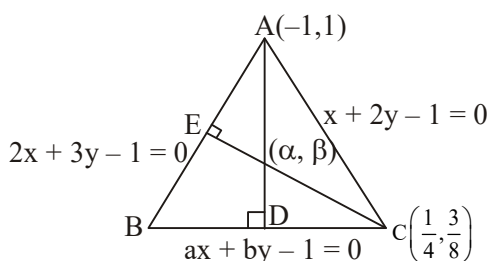
$$\therefore \alpha^2 + \gamma^2 = 14$$

37. If the orthocentre of the triangle formed by the lines $2x + 3y - 1 = 0$, $x + 2y - 1 = 0$ and $ax + by - 1 = 0$, is the centroid of another triangle, whose circumcentre and orthocentre respectively are $(3, 4)$ and $(-6, -8)$, then the value of $|a - b|$ is _____.

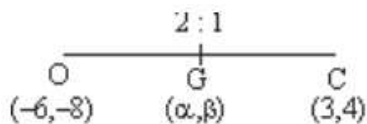
यदि रेखाओं $2x + 3y - 1 = 0$, $x + 2y - 1 = 0$ और $ax + by - 1 = 0$ द्वारा बनाया गया त्रिभुज, जिसका लंब केन्द्र किसी अन्य त्रिभुज के केन्द्रक के बराबर है एवं जिसका परिकेन्द्र और लंब केन्द्र क्रमशः $(3, 4)$ तथा $(-6, -8)$ है, तो $|a - b|$ का मान होगा _____

Ans. 16

Sol.



For another Δ



$$\alpha = \frac{2 \times 3 - 6}{3} = 0 \quad \beta = \frac{2 \times 4 - 8}{3} = 0$$

$$(\alpha, \beta) = (0, 0)$$

$$x + 2y - 1 = 0$$

$$2x + 3y - 1 = 0$$

$$\begin{array}{r} - - + \\ \hline y - 1 = 0 \end{array}$$

$$y = 1$$

$$x = -1$$

$$A(-1, 1)$$

$$\text{Slope of AD} = \frac{0 - 1}{0 + 1} = -1$$

and

$$M_{AD} \times M_{BC} = -1$$

$$(-1) M_{BC} = -1$$

$$M_{BC} = 1$$

Now for point C

find equation of CE

$$M_{CE} = \frac{2}{3}$$

$$y = \frac{2}{3}x$$

and equation of AC

$$x + 2y - 1 = 0$$

$$\underline{-3x + 2y = 0}$$

$$4x - 1 = 0$$

$$x = \frac{1}{4}$$

$$y = \frac{3}{2} \times \frac{1}{4} = \frac{3}{8}$$

Eq. of BC

$$C\left(\frac{1}{4}, \frac{3}{8}\right)$$

$$m_{BC} = 1$$

$$y - \frac{3}{8} = 1\left(x - \frac{1}{4}\right)$$

$$\frac{8y - 3}{8} = \frac{4x - 1}{4}$$

$$8y - 8x + 2 - 3 = 0$$

$$-8x + 8y - 1 = 0$$

$$ax + by - 1 = 0$$

$$a = -8$$

$$b = 8$$

$$\text{Value of } |a - b| = |-8 - 8| = 16$$

Circle

38. If the four distinct points $(4, 6)$, $(-1, 5)$, $(0, 0)$ and $(k, 3k)$ lie on a circle of radius r , then $10k + r^2$ is equal to :

यदि चार भिन्न बिन्दु $(4, 6)$, $(-1, 5)$, $(0, 0)$ और $(k, 3k)$, त्रिज्या r के एक वृत्त पर हैं, तो $10k + r^2$ बराबर है :

- (A) 34 (B) 33 (C) 32 (D) 35

Ans. D

Sol. $x^2 + y^2 + 2gx + 2fy = 0$

$$16 + 36 + 8g + 12f = 0 \Rightarrow 2g + 3f + 13 = 0 \quad \dots(1)$$

$$1 + 25 - 2g + 10f = 0 \Rightarrow 2g - 10f - 26 = 0 \quad \dots(2)$$

From (1) - (2)

$$13f + 3g = 0 \Rightarrow f = -3$$

$$\text{Also } 2g - 9 + 13 = 0 \Rightarrow g = -2$$

$$\text{So circle } \Rightarrow x^2 + y^2 - 4x - 6y = 0 \Rightarrow k^2 + 9k^2 - 4k - 18k = 0$$

$$10k = 22$$

$$\text{So, } 10k + r^2 = 22 + 9 + 4 = 35$$

39. Let C_1 be the circle in the third quadrant of radius 3, that touches both coordinate axes. Let C_2 be the circle with centre $(1, 3)$ that touches C_1 externally at the point (α, β) . If $(\beta - \alpha)^2 = \frac{m}{n}$, $\gcd(m, n) = 1$, then $m + n$ is equal to :

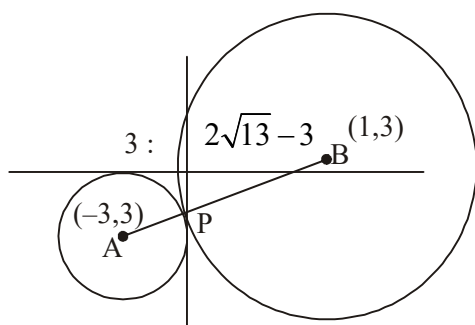
माना तृतीय चतुर्थांश में त्रिज्या 3 का एक वृत्त C_1 है, जो दोनों निर्देशांक अक्षों को स्पर्श करता है। माना केन्द्र $(1, 3)$ का एक वृत्त C_2 है,

जो वृत्त C_1 को बाह्यतः बिन्दु (α, β) पर स्पर्श करता है। यदि $(\beta - \alpha)^2 = \frac{m}{n}$, $\gcd(m, n) = 1$, तो $m + n$ बराबर है :

- (A) 9 (B) 31 (C) 13 (D) 22

Ans. D

Sol. $PB = \sqrt{16 + 36}$
 $= \sqrt{52} = 2\sqrt{13}$



$$\alpha = \frac{3-3(2\sqrt{13}-3)}{2\sqrt{13}}, \beta = \frac{9-3(2\sqrt{13}-3)}{2\sqrt{13}}$$

$$(\beta - \alpha)^2 = \frac{36}{4 \times 13} = \frac{9}{13} = \frac{m}{n}$$

So

$$m + n = 22$$

40. Let circle C be the image of $x^2 + y^2 - 2x + 4y - 4 = 0$ in the line $2x - 3y + 5 = 0$ and A be the point on C such that OA is parallel to x-axis and A lies on the right hand side of the centre O of C. If $B(\alpha, \beta)$, with $\beta < 4$, lies on C such that

that the length of the arc AB is $\left(\frac{1}{6}\right)^{\text{th}}$ of the perimeter of C, then $\beta - \sqrt{3}\alpha$ is equal to :

माना कि वृत्त C, $x^2 + y^2 - 2x + 4y - 4 = 0$ का रेखा $2x - 3y + 5 = 0$ में प्रतिबिम्ब है और A, C पर एक बिन्दु इस प्रकार है कि OA, x-अक्ष के समान्तर है और A, C के केन्द्र O के दाहिनी ओर स्थित है। यदि $\beta < 4$ के साथ $B(\alpha, \beta)$ पर इस प्रकार स्थित है कि

चाप AB की लम्बाई C की परिधि की $\left(\frac{1}{6}\right)$ है, तो $\beta - \sqrt{3}\alpha$ बराबर है :

(A) $4 - \sqrt{3}$

(B) $3 + \sqrt{3}$

(C) 3

(D) 4

Ans. D

Sol. Centre $(1, -2)$, $r = \sqrt{1+4+4} = 3$
Image of $(1, -2)$ in $2x - 3y + 5 = 0$ is

$$\frac{x-1}{2} = \frac{y+2}{-3} = \frac{-2(2+6+5)}{4+9}$$

$$\frac{x-1}{2} = \frac{y+2}{-3} = -2$$

$$x = -3, y = 4$$

$$\therefore \text{Equation of C : } (x+3)^2 + (y-4)^2 = 9$$

$$A(-3+3\cos\theta, 4+3\sin\theta), O(-3, 4)$$

$$m_{OA} = \frac{3\sin\theta}{3\cos\theta} = 0 \Rightarrow \theta = 0$$

$$A(0, 4)$$

$$P = 2\pi r = 6\pi$$

$$L \text{ of } A, \ell = r\theta$$

$\Rightarrow$

$$\pi = 3\theta$$

$$\theta = \frac{\pi}{3}$$

$$B\left(-3 + \frac{3}{2}, 4 - \frac{3\sqrt{3}}{2}\right)$$

$$B\left(-\frac{3}{2}, 4 - \frac{3\sqrt{3}}{2}\right)$$

$$\therefore \beta - \sqrt{3}\alpha = 4 - \frac{3\sqrt{3}}{2} + \frac{3\sqrt{3}}{2} = 4$$

